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Intel RSTe AHCI RAID Driver 4.5.10.1021 on Supermicro X9QRI-F+ Motherboard

1. Overview and Architecture

Intel Rapid Storage Technology Enterprise (RSTe) version 4.5.10.1021 is a comprehensive software RAID stack designed for Intel server chipsets. It provides both firmware (pre-boot) and OS-level components to manage storage arrays on Intel C600 series platforms (e.g. the X9QRI-F+'s C602 chipset). The RSTe architecture includes:
- **Pre-Boot RAID Firmware:** Legacy BIOS option ROM and UEFI drivers for both the SATA AHCI controller and the SCU (Storage Control Unit) controller [1] [2]. These enable RAID configuration and boot support before any OS loads (e.g. via a BIOS Ctrl+I utility for legacy mode or UEFI RAID driver).
- **OS Drivers:** Two distinct driver modules are used in Windows – one for the SATA AHCI controller (iaStorA) and one for the SATA/SAS SCU controller (iaStorS). The AHCI driver manages drives on the standard SATA ports (in either AHCI pass-through or RAID mode) [3], while the SCU driver manages devices on the SAS/SATA capable ports (providing pass-through or full RAID functionality for those ports) [4]. On the X9QRI-F+, the C602 PCH's six onboard SATA ports (2 × SATA3 6Gb/s + 4 × SATA2 3Gb/s) would be handled by the AHCI driver; if any additional SAS ports were present via the SCU, they'd be handled by the SCU driver. (In practice, the X9QRI-F+ exposes 6 SATA ports from the C602 chipset, and no separate SAS expander on this board, so RAID is confined to those SATA ports.)
- **Management Tools:** RSTe includes a Windows-based **GUI utility**, a **Command-Line Interface (CLI)** utility, and a **CIM** (Common Information Model) plugin for management [5]. The GUI (Intel RSTe management console) is a graphical application for creating and monitoring RAID volumes on the connected drives [6]. The CLI (`RstCli64.exe`) offers text-based commands to perform RAID operations (create, delete, query arrays) without a GUI – useful for scripts or Server Core installations [7]. The CIM component is a standards-based interface (conforming to SMIS – Storage Management Initiative Specification) that allows enterprise management software to query and control the RAID setup via a CIM/WMI provider [8]. This means administrators can integrate RSTe management into broader systems management tools through an industry-standard API.

Feature Set: As an enterprise-oriented RAID solution, Intel RSTe supports a rich feature set beyond basic RAID levels. It supports RAID 0, 1, 5, 10 arrays on the chipset's SATA/SAS controllers [9], including the ability to mix SAS and SATA drives in RAID (when using an appropriate chipset and license key). Notable features include support for **SAS drives and expanders**, hot-plug/hot-swap drive support with active I/O, **hot spare** disk assignments and automatic rebuild, and online RAID level migration and expansion (Online Capacity Expansion) [10] [11]. RSTe implements enterprise-grade reliability features such as **dirty stripe journaling** and **Partial Parity Logging (PPL)** to improve RAID5 recovery after unexpected outages [12] [13]. It also supports **NCQ** for SATA (and command queuing for SAS) to optimize performance [14], background **patrol reads** for data scrub, SMART monitoring, and SGPIO drive enclosure LED management [15] [16]. The firmware/driver stack handles **auto-rebuild on hot-insert**, configurable stripe sizes, and provides email alerting for critical events [17]. In summary, the RSTe 4.5 driver package is a hybrid software RAID solution (sometimes called "firmware RAID") that leverages the CPU to implement RAID but uses the chipset's integrated controllers and firmware hooks to provide a seamless RAID experience in enterprise environments.

2. Compatibility Matrix (OS, Chipset, Hardware)

The **Intel_RSTE_AHCI_Win_Drivers_GUI_CLI_CIM_4.5.10.1021.zip** package is targeted at Intel's server platforms from roughly the **2011–2014 era**. Specifically, it supports **Intel 60x/C600 series chipsets** (e.g. C602, C604, C606) and their successors in the C610, C220, and C230 series [18]. These chipsets correspond to platforms using Xeon E5-2600 v1/v2 (C600 series) and Xeon E5 v3/v4 (C610 series), as well as certain Xeon E3 platforms (C220/C230 for Xeon E3 v3/v4). The Supermicro **X9QRI-F+**, with its Intel C602 PCH, falls squarely in the supported chipset list. (Intel's own S2600 family server boards with C602/C604 chipsets are explicitly listed as supported by this driver package [19] [20], which implies any OEM board using the same chipset – like Supermicro's X9 series – is compatible.)

Supported Operating Systems: According to Intel's documentation, RSTe 4.5.10.1021 supports a range of Windows OSes common in that timeframe. This includes client OS versions and server editions: **Windows 7** and **Windows 8.1** (and their 64-bit variants), **Windows 10** (up to the 2018 release), as well as **Windows Server 2008 R2**, **Windows Server 2012**, and **Windows Server 2012 R2** [21]. (Softpedia's release notes confirm the same, listing Windows 7/8.1/10 and Server 2008 R2/2012/2012R2 as officially supported platforms [22].) Notably, **Windows Vista and Windows Server 2003** are *not* supported by this 4.5 driver; older RSTe releases would be required for those legacy OSes (Intel notes that the "3.7 release does not support Vista – use a prior release for Vista" [23]). Additionally, Windows Server 2016/2019 are outside the scope of 4.x drivers – support for those came with later RSTe 5.x/6.x versions. Linux systems are not managed by this Windows driver, of course; Linux has its own MD-RAID or uses the Intel RSTe RAID metadata via the kernel's IMSM support if applicable.

Supported Hardware & RAID Configurations: The driver supports both **AHCI-capable SATA controllers and the integrated SATA/SAS controllers** (SCU) present on these chipsets [18]. In practice, on platforms like the X9QRI-F+ that have six SATA ports from the chipset, RSTe can manage those ports in either straight AHCI mode (drives exposed individually to OS) or RAID mode. When RAID mode is enabled in BIOS, you can create RAID volumes across those SATA drives. If the platform had an SCU with SAS ports (for example, some motherboards have additional SAS ports via the chipset – often requiring a hardware key to unlock), RSTe would manage those as well, enabling RAID with SAS drives. The **Intel RAID C600 Upgrade Key** is an accessory for certain boards to unlock full SAS RAID capabilities; with RSTe 4.5, if that key is present, the chipset's SAS controller can be used (and SAS-specific features become available) [24]. It's important to note that RSTe is **specific to Intel chipset storage** – it won't manage third-party storage controllers. On the X9QRI-F+, this means RSTe will handle the onboard Intel SATA ports; any additional storage controllers (e.g. an LSI/AVAGO PCIe RAID card) would use their own drivers.

Device and Volume Limits: RSTe (as of 4.5) supports up to 8 drives on the SCU SAS controller and 6 drives on the AHCI SATA controller of C600/C610 chipsets (matching the physical ports available). RAID arrays can be created with 2 to 8 drives depending on level (RAID 0/1/10/5 supported). There are a few constraints: for example, **only 4-drive RAID10 is supported** (striped mirror using four disks) [25], and larger RAID10 sets (spanning more drives) were not validated in this release. RAID5 is supported *on pure SATA* setups, but **if SAS features are enabled via upgrade key, RAID5 is disabled on that controller** [24] (likely because enabling "enterprise" SAS mode hands off RAID5 to hardware RAID solutions or to encourage use of Intel's premium RAID key). Also, some niche configurations are unsupported – e.g. creating a single SAS "wide port" that spans both SCU controllers is not allowed [26], and you cannot attach both SCU channels to the same SAS expander backplane [27]. These limitations are more relevant to complex SAS backplane configurations than to a simple 4U chassis with direct SATA connections.

3. GUI, CLI, and CIM Components Description

GUI (Graphical User Interface): The RSTe package includes a Windows GUI application (often simply called **Intel Rapid Storage Technology enterprise console**). This is a user-friendly management console that runs in Windows, providing real-time status of connected drives and RAID volumes. Through the GUI, an administrator can create new RAID volumes (choosing the RAID level, member disks, stripe size, etc.), delete or modify volumes, assign hot spares, and monitor the health/status of arrays. The GUI also surfaces events and can be configured to trigger **email alerts** for drive failures or array degradation (it uses the Intel Accelerated Storage Manager service in the background to handle such tasks) [17] [28]. On launching the GUI (after installing the "Drivers_GUI_CLI_CIM" package and required prereqs), the tool will show the Intel RSTe branding and list all drives attached to the chipset's controllers. It only manages the Intel AHCI/SCU controllers – for example, drives on a separate HBA wouldn't appear. The interface typically requires the **.NET Framework** (v3.5 in this era) to be present on the system for the GUI to function [29]. On a fresh Windows Server installation (especially 2008 R2/2012), .NET 3.5 might need to be enabled prior to installing the RSTe GUI. Once running, the GUI offers a convenient way to supervise the RAID without rebooting into BIOS; one can initiate rebuilds or verify tasks from within Windows, and the tool will show progress of mirror rebuilds or parity verifications. For the X9QRI-F+ motherboard, system administrators can rely on this GUI to manage the onboard RAID sets during normal operation.

CLI (Command Line Interface): For scripting or environments without a GUI (e.g. Windows Server Core or for automated deployment), Intel provides the **RSTCLI** utility. In this driver package, it's typically included as `rstcli.exe` (32-bit) and `rstcli64.exe` (64-bit), found in the "CLI" folder after extraction. The RSTe CLI allows **basic RAID management via command line** [7]. It supports commands to **create** arrays, delete arrays, query information (controllers, disks, volumes, enclosures), manage rebuilds, and modify arrays (for example, expanding a volume or adding a spare) [30] [31]. The CLI syntax is fairly straightforward, and typing `rstcli --help` will list available modes and options [32]. An example operation might be: `rstcli --create --level 1 --name "OS_Mirror" 0 1` (which would create a RAID1 named OS_Mirror on disks 0 and 1). The CLI communicates with the same underlying driver services as the GUI, so any changes made via CLI will be reflected in the GUI and vice versa. This tool is particularly useful on the Supermicro X9QRI-F+ if running a non-GUI server environment – for instance, Hyper-V Server or Windows Server Core – where you still need to manage the RAID. It's also handy for writing batch files to automate RAID configuration during system provisioning. The RSTCLI does *not* depend on the OS being fully up – it can even be used in the pre-boot environment (Intel provides UEFI versions of the CLI for use in the EFI shell as well) [33], although typical usage on Windows is within the running OS. According to Intel's documentation, *"RSTCLI is an end user command line utility used to do basic RAID operations on Intel RSTe enabled systems… supports RAID0, RAID1, RAID5 and RAID10 volumes"* [34], covering the common needs for array setup and maintenance.

CIM (Common Information Model) Plugin: The third component, often less visible to end-users, is the CIM Provider. This is essentially a background service/plugin that exposes the RAID management to enterprise management frameworks via standardized CIM classes (per the DMTF SMIS storage management spec) [8]. In simpler terms, the CIM component allows software like monitoring agents or external management consoles (for example, a data center management tool, or Windows WMI scripts) to query the RAID controller status and perform management actions through a well-defined interface, rather than using Intel's proprietary GUI or CLI. Intel's RSTe CIM implementation provides a subset of profiles such as the "Host Hardware RAID Controller" profile under SMIS [35]. System integrators might leverage this to integrate RAID health checks into their centralized dashboards – e.g., checking via WMI if an array is degraded, or using PowerShell/WBEM to get drive info. On Windows, the CIM provider likely appears as a service or driver that registers WMI classes representing the RAID arrays. While not usually interacted with directly by admins, it's important for integration – for example, Supermicro's management software or third-party tools could use the CIM interface to raise an alert in case of a drive failure. In summary, the CIM plugin is there to ensure the RSTe-managed storage can be **monitored and controlled programmatically** in enterprise environments using standard protocols, complementing the human-facing GUI/CLI tools.

4. Performance Impact and Benchmarks

Intel RSTe is a software RAID solution that operates using the system CPU, so performance can vary depending on workload and system resources. In practice, RSTe has been tuned to provide significant improvements in I/O efficiency over generic AHCI drivers or older firmware-RAID implementations. Intel notes that using the RST family drivers can yield **improved storage performance, lower power consumption, faster boot times, and quicker data reads** compared to default OS drivers [36]. These gains come from features like NCQ queue optimization, write-back caching (for RAID1/5 with battery/capacitor backup, if enabled), and driver-level optimizations. For example, RAID0 arrays under RSTe can approach the linear throughput of the combined drives (limited by the DMI bus and CPU overhead). RAID1/10 read performance is enhanced by reading from multiple mirrors in parallel. The driver also offloads some background tasks (like rebuild verification and patrol reads) to idle cycles, which helps maintain performance during array maintenance.

Known Performance Characteristics: Early versions of RSTe (3.x series) had a reputation of slightly lower performance compared to Intel's desktop RST drivers, but by the 4.x series many of those gaps closed. The RSTe 4.5 driver is optimized for high-concurrency enterprise workloads. It supports **Native Command Queuing** deeply, which is beneficial for SSDs and fast HDDs under queue depth. With SAS drives (on chipsets where SAS is enabled), RSTe can use Tagged Command Queuing similarly. One feature, **Partial Parity Logging (PPL)**, improves RAID5 write performance in crash scenarios by avoiding a full stripe rebuild after an unclean shutdown – instead only the partial stripe needs re-sync [37], which indirectly means better effective performance (since recovery time is shorter, less performance degradation after power loss).

There aren't official public benchmarks in the Intel documentation for RSTe 4.5 specifically, but anecdotal reports and previous testing show that for RAID0 and RAID1, performance is on par with or exceeding typical hardware SATA RAID controllers. RAID5 performance under RSTe will depend on CPU horsepower – on a multi-core Xeon, the parity calculations are usually not a bottleneck for moderate I/O, but under very heavy random write loads a dedicated RAID card with XOR offload might outperform it. That said, RSTe allows leveraging modern CPU vector instructions for parity, so even software RAID5 can be quite efficient. Intel at one point advertised improved IOPS in certain 4.x updates – e.g., a mention of **"Intel improves 285K IOPS performance with a big update"** was noted in community discussions (referring to NVMe RAID, which is tangential to this AHCI driver) – demonstrating ongoing performance tuning. On the X9QRI-F+ (with up to four Xeon E5-4600 CPUs available), the available processing power is ample to handle RAID overhead, meaning the storage subsystem likely runs at close to the maximum speed of the SATA 6G links for sequential workloads.

Impact on the Storage Subsystem: When enabled, RSTe's RAID mode replaces the default AHCI driver. There is some CPU overhead – e.g. a few % CPU utilization during heavy array rebuild or verification – but in return you gain fault-tolerance or performance benefits of RAID. In most cases, the impact is minimal on a server-class system; the driver operates in kernel mode efficiently. One consideration is that RSTe's write-back cache (for RAID5 or RAID1 write cache) should ideally be used with a UPS or backup power, as an unexpected power loss could otherwise risk data in-flight (the driver does use journaling/PPL to mitigate this risk [12]). Latency-wise, running in RAID mode vs AHCI pass-through is almost identical for reads, and slightly higher for writes (due to parity or mirroring overhead). The benefit is that with RAID0 you get aggregated throughput, with RAID1/10 you get redundancy (and improved read speeds in RAID10), and with RAID5 you get a balance of capacity and redundancy at the cost of some write overhead. Intel RSTe is generally considered robust for enterprise use – it was validated on server workloads, meaning it can handle continuous heavy I/O (databases, Hyper-V VMs, etc.) reliably. If absolute maximum performance with zero CPU overhead is required, a dedicated RAID card might be chosen instead, but for many integration scenarios the RSTe driver offers an excellent trade-off: near-hardware RAID performance with the flexibility of software RAID.

5. Known Issues, Bugs, and Security Concerns

Throughout its release cycle, the RSTe 4.x line has addressed various bugs and carries a few **known limitations** as documented by Intel. The **release notes for version 4.5.10.1021** (and its immediate predecessors) highlight several fixes: Version 4.5.10.1021 itself included installer fixes to resolve installation glitches in certain scenarios [38]. The prior update (4.5.2.1011) fixed issues with RAID array creation when using 7 or 8 drives, resolved problems with the CIM provider installation, and updated the supported chipset list (ensuring newer chipset IDs were recognized) – these addressed cases where large drive counts or management components could fail to work initially. Earlier 4.x versions also incorporated fixes like support for Windows 10 and NVMe devices (more on NVMe in section 6), as well as stability improvements for suspend/resume and power management corner cases.

Known Issues/Restrictions: The RSTe 4.5 driver has a list of known restrictions which can be considered "open issues" or design limitations (some of which were noted earlier): for example, when the Intel RAID C600 SAS upgrade key is used, RAID5 is disabled on that controller [24]. Only four-drive RAID10 is supported (larger RAID10 arrays were not validated) [25]. Multi-path I/O (MPIO) load balancing is not supported in RAID mode [26] – meaning if you have a dual-path SAS enclosure, only one path is used by RSTe. Certain specific SAS drives (notably some WD 6Gbps SAS models in the S25/S35 family) are not supported and may not be recognized or may error [27]. The BIOS POST messages for RAID (during boot) only appear if more than one drive is attached – so a single-drive in RAID mode might not show a RAID BIOS screen [39]. There is also an LED behavior quirk: during a RAID migration or rebuild, not all drive fault/activity LEDs may blink correctly to indicate rebuild status [40]. If a system crashes or loses power under heavy RAID I/O, an **manual verify/repair** might be required on reboot to ensure consistency [41] – RSTe will usually prompt for or automatically start this check. Additionally, RAID volume name length is limited to 15 characters (16+ will be truncated or unsupported) [42]. None of these are showstoppers for operation, but they are useful to note for administrators to avoid confusion (for instance, if you see the volume name truncated, or the LEDs not blinking during a rebuild, it's a known behavior).

Security Concerns: In early 2019, a security vulnerability was disclosed in the RSTe software – specifically in the **Intel Accelerated Storage Manager (ASM)** service which underpins the RSTe GUI. The issue (INTEL-SA-00231, CVE-2019-0135) was an **elevation of privilege** vulnerability in the RSTe installer/management service, where improper default permissions could allow a local authenticated user to gain higher privileges [43]. This affected RSTe versions **before 5.5.0.2015**. Since our focus is on 4.5.10.1021, it means this version is indeed impacted by that vulnerability. Intel's guidance was to update to RSTe 5.5 or later to patch the hole [44]. In context, on a Supermicro X9QRI-F+ (a server likely sitting in a data center), this vulnerability would mean that if an unprivileged user had local login or the ability to execute code, they might leverage the RSTe service to escalate privileges on that system. As such, if this platform is being used in a sensitive environment, one should restrict access to trusted admins only, or consider upgrading to a fixed RSTe release if available for that chipset (though official 5.x support for C602 might not exist, in which case mitigating by locking down the service permissions is wise). Other than that specific CVE, there haven't been major security issues publicized for RSTe 4.5 – but, as with all drivers, it's advisable to keep systems updated (within what the hardware supports) to pick up any stability and security improvements.

Notable Bug Reports: Most user-reported issues with RSTe 4.5.x revolve around installation and monitoring rather than catastrophic failures. For instance, some workstation users noted confusion that the package installs driver version 4.5.0.1234 while the installer is 4.5.10.1021 – this is expected (the package version vs driver binary version differ slightly) [45]. Another common point of confusion is mixing RSTe with consumer RST drivers; on X9 series boards, only RSTe is supported – attempting to use an Intel RST (non-enterprise) driver can fail. There was also an Intel community discussion about RSTe on newer OS (e.g. trying to use it on Server 2016) where the RSTe GUI wouldn't run due to .NET issues or missing support – reinforcing that one should stick to the validated OS list or use newer RSTe versions for newer OS [46]. Overall, the 4.5.10.1021 driver is considered **stable for production** on the intended platforms, with the known limitations discussed above.

6. Change History and Version Evolution

Intel RSTe underwent several iterations leading up to 4.5.10.1021. Here's a brief history of how this version differs from prior releases:

- **Earlier 3.x Series:** RSTe 3.0 through 3.8 (circa 2012–2013) were the initial releases for the C600 chipsets. They introduced core RAID features and supported Windows 7/8 and Windows Server 2008 R2/2012. Over the 3.x updates, Intel added things like support for drive LED indicators via SGPIO backplanes and even retroactive support for legacy OS (e.g. one 3.x update added **Windows Server 2003 R2 SP2 (32-bit)** support for certain configurations [47]). By 3.8, features like TRIM on RAID0 (for SSDs) and better stability were incorporated. However, Windows 10 was not yet supported in those versions, and the GUI was based on older frameworks.

- **Introduction of 4.x (PV release):** The 4.x "Production Version" releases of RSTe coincided with Intel's Grantley platform (Xeon E5 v3, C610 chipset) around 2014–2015. One of the big additions in RSTe 4.0+ was official support for **Windows 10** and newer server OSes, and a unification of the driver for C600 and C610 families. For example, **RSTe 4.5.0.2123** in 2016 was notable for introducing support for **NVMe RAID** (branded as RSTe NVMe) for data center SSDs [48] [49]. In fact, Intel split off an "RSTe NVMe" driver package at that time, because NVMe RAID on the enterprise platform was added – this was a separate driver focusing on NVMe drives (as opposed to the AHCI/SAS driver for the chipset SATA/SAS ports) [50]. The 4.5.0.x branch around 2016 was a significant evolution: NVMe RAID support (non-bootable initially) for SSD 750 and DC P3xxx series, a rename of the NVMe driver components, etc. [51] [52].

- **Evolution to 4.5.10.1021:** The version in question (4.5.10.1021 dated mid-2018 [53]) is one of the later updates in the 4.x line. According to Intel, by that point the **Intel C600/C220 series chipsets were considered End-of-Life** in terms of adding new features [54] – the 4.5 driver was essentially providing maintenance updates for those older platforms while newer platforms (Xeon Scalable, etc.) moved to RSTe 5.x and beyond. Changes in the 4.5.x series mostly involved bug fixes and minor enhancements: for example, **4.5.1 and 4.5.2** releases fixed the installation and drive count issues as noted earlier, and refined the support for large configurations. The jump to 4.5.10 likely indicates cumulative minor updates (installer refinements, perhaps signed driver updates for Windows 10 compatibility). It's worth noting that the driver inside is version 4.5.0.1234, which implies the core driver hasn't drastically changed – it's the packaging/installer that saw incrementing versions. Intel's provided README for 4.5.10 lists "Installer fixes" as the primary change [38], suggesting that previous releases had some installer or upgrade hiccups that were resolved.

- **Post-4.5 (5.x Series):** After 4.5.x, Intel's next major step was RSTe 5.0/5.5 which were aligned with the Purley platform (Xeon Scalable, C620 series chipset) and also backported support for Windows Server 2016/2019. In doing so, they **dropped official support for the older C600/C610 in new releases** – the Intel release notes indicate that after RSTe 4.5, those earlier chipsets were no longer in the support matrix [55]. This means 4.5.10.1021 is effectively the *final* recommended version for boards like the X9QRI-F+ with C602. Users of such boards should stick to 4.5.10 (or the slightly earlier 4.5.x versions) for best results, as the 5.x drivers may not recognize the older hardware. The 5.x line also addressed the aforementioned security issue (by version 5.5.0.2015 in 2019) [44] and improved features like bootable NVMe RAID. But those developments are mostly beyond the scope of the X9 platform usage.

In summary, version 4.5.10.1021 doesn't introduce major new features over earlier 4.x builds, but it consolidates all prior fixes making it the most stable release for the X9QRI-F+ era hardware. It ensures compatibility with the latest OS that the board is likely to run (Windows 10 or Server 2012 R2), and is the end-of-line update for that generation.

7. Deployment on Supermicro X9QRI-F+ Platform

The Supermicro **X9QRI-F+** is a quad-socket Sandy/Ivy Bridge-EP server motherboard with the Intel C602 chipset. In the context of storage and RAID, here is how RSTe 4.5.10.1021 applies to this board:

- **Onboard SATA RAID:** The X9QRI-F+ provides 6 onboard SATA ports via the C602. Two of these are 6 Gb/s (SATA3) and four are 3 Gb/s (SATA2) [56] [57]. Out of the box, these ports can be run in AHCI mode (no RAID, each drive independent) or in Intel RSTe RAID mode. The board's BIOS Setup has an option (likely under SATA Configuration) to choose RAID Mode for the PCH SATA controller. When set to RAID, the Intel RSTe firmware takes over those ports. This enables the RAID BIOS interface (accessible via **Ctrl+I** during POST with the legacy BIOS, or via a BIOS menu/UEFI driver on newer firmware) to create RAID volumes that the system will see as single bootable drives. The RSTe 4.5 driver must then be loaded in the OS to recognize and manage these RAID volumes.

- **Driver Installation:** For Windows installations on this platform, if you plan to boot from a RAID volume, you'll need to supply the RSTe driver during OS installation (especially for older Windows versions that may not have a built-in driver). The Intel download "GUI_CLI_CIM_4.5.10.1021.zip" contains both the driver INF files and an installer. Supermicro typically would provide a download or include the driver on their support DVD as well. The installation steps would be: enable RAID in BIOS, configure your array in the Intel RAID BIOS, then when installing Windows, use "Load Driver" to load the `iaStorA.inf` / `iaStorS.inf` from the RSTe package (or run the installer after installation for a data drive). The 4.5.10.1021 package supports Windows 7/8/10 and Server 2008R2/2012(R2), so whichever OS the X9 board is running, this package covers it [58]. After OS setup, installing the full package via its **Setup.exe** will also install the management GUI and other components.

- **Relevance of RSTe on X9QRI-F+:** In an integration scenario, one might ask whether to use the onboard RSTe RAID or a hardware RAID card. The answer depends on needs: The onboard RSTe RAID is quite capable for many applications – for example, a mirrored boot drive or a RAID5 for a data volume can be achieved without extra cost. The X9QRI-F+ being a high-end board (quad CPU) implies it might be used in high-performance servers where often a dedicated RAID HBA with cache is used for heavy I/O. However, RSTe provides a good option for those who want to use the chipset's capabilities (for instance, creating a RAID10 across four SATA SSDs for a scratch database). It's also useful in environments where an HBA failed – as a temporary measure, the onboard RSTe could rebuild an array. From a manageability standpoint, because the X9QRI-F+ supports IPMI (lights-out management), one can potentially monitor the RAID status via IPMI events if the Supermicro IPMI firmware hooks into the RSTe (though typically IPMI doesn't directly know about RSTe volumes without additional software).

- **RAID Key and SAS Usage:** The X9QRI-F+ does **not** have onboard SAS ports (only the 6 SATA). Some other X9 boards (like those with suffix "-TF+" including an LSI controller or with an add-on SAS mezzanine) might have different RAID options. Since X9QRI-F+ is SATA-only, the Intel RAID C600 Upgrade Key (for SAS) is not applicable – there are no SCU SAS connectors to unlock. So, one benefit is that **RAID5 is supported on this board** using RSTe (the limitation of RAID5 being disabled only applies if SAS is enabled, which it isn't here) [24]. That means an admin could create a RAID5 across, say, three or four SATA drives on the X9QRI-F+ if desired (keeping in mind the performance considerations of RAID5).

- **Use in Practice:** Many users of X9 boards have reported success using RSTe for simple RAID setups, like two SSDs in RAID1 for the OS. The driver's maturity by version 4.5 means that on Windows Server the arrays are stable and rebuild predictably if a drive is replaced. It's important when deploying to have some strategy for monitoring – e.g., configure email alerts in the RSTe GUI or use Supermicro's SuperDoctor software (if it can interface with RSTe events). Because if a drive in a RAID1 fails, you'd want to be notified (the system will continue running, as the array is redundant, so without monitoring it could go unnoticed until the second drive fails). The RSTe driver will log events to the Windows System Event Log (with source IAStorA/IAStorS), which can be caught by monitoring tools.

- **Firmware/Bios Dependencies:** Ensure the X9QRI-F+ is running a relatively recent BIOS that supports Intel RSTe Option ROM version corresponding to 4.x. Supermicro did release BIOS updates over the life of X9 boards that sometimes updated the RAID firmware. Even if the BIOS has an older RAID ROM (e.g. 3.x), the 4.5 OS driver should still work – Intel designs them to be backward compatible – but some advanced features might not be available until the Option ROM is updated. If there's a BIOS setting for "Intel RSTe RAID" vs "LSI ESRT2" (some boards have a choice for the SCU controller mode), on this board with only SATA you won't see that – it's always RSTe for the PCH SATA RAID.

In summary, deploying RSTe 4.5.10.1021 on the Supermicro X9QRI-F+ allows you to utilize the board's integrated storage controller to set up reliable RAID arrays. This is highly relevant for hardware integration engineers who want to avoid adding PCIe RAID cards and use existing chipset features. The key tasks are enabling RAID in BIOS, configuring the array, loading the driver during OS install, and then using the GUI/CLI to manage the array going forward.

8. Configuration Notes and BIOS/UEFI Considerations

When working with the X9QRI-F+ and RSTe, keep in mind these configuration and firmware details:

- **BIOS Setting:** In the BIOS Setup Utility of the X9QRI-F+, the SATA mode should be set to **"RAID" (Intel RSTe)** for the ports that will be part of RAID. Typically, the BIOS will list options like AHCI or RAID for the PCH SATA. Select RAID before installing the OS. If the system was already installed in AHCI mode, switching to RAID mode without reinstallation is possible (by pre-loading the RSTe driver in the OS), but it's simpler to decide upfront.

- **Option ROM vs UEFI mode:** The X9 generation boards primarily boot via Legacy BIOS by default. In Legacy mode, after enabling RAID, you will see the **Intel RAID Option ROM** during POST (usually showing a message about pressing Ctrl+I to configure RAID). Enter that interface to create your volumes. This is a text-mode BIOS utility provided by the RSTe firmware [1] [59]. If you are using UEFI Boot (and have CSM disabled), the process is slightly different: the board uses Intel's UEFI driver for RAID, and configuration can be done either through a UEFI shell using the RSTe CLI or possibly through a BIOS advanced menu for storage. Many admins find it easier to temporarily enable legacy mode, do Ctrl+I to set up RAID, then continue with UEFI OS installation. Either way, ensure that the boot order is set such that the RAID volume (often presented as "Intel Volume_0000" or the name you gave it) is the first boot device after creation.

- **Driver Loading:** For Windows Server 2008 R2/2012, etc., you will almost certainly need to load the RSTe driver during OS installation if your boot drive is on the RAID. Windows 10/Server 2012 R2 might have an inbox driver that recognizes the RAID volume (sometimes it shows up as a single disk without additional drivers), but it's recommended to use Intel's driver for full feature support. The driver INF can be found under a folder in the package (e.g., `Drivers\64-bit\` etc.). Supermicro might provide a floppy image or USB drivers for this purpose as well.

- **Microsoft .NET Framework:** As mentioned, the RSTe GUI depends on .NET (v3.5 for older OS, or it might work with .NET 4.x on newer OS). On **Windows Server 2012 R2/2016**, the installer should auto-enable the required .NET components if connected to the internet or installation media. On **Windows Server 2008 R2**, you may need to manually enable .NET 3.5 (as it's a Feature in Server Manager) before installing the RSTe GUI [29]. If the GUI fails to launch, check that .NET is present.

- **Caching and Write-back Settings:** The RSTe GUI allows toggling the write cache policy on RAID volumes. By default, RAID1 and RAID5 might be set to write-through (safer, but slower). If you have a UPS on the system, you might consider enabling write-back cache on the volume for better performance – but note that this *could* risk data on power loss (though RSTe's journaling mitigates it somewhat). The Option ROM or UEFI utility might also have a setting for "Cache Enabled = Yes/No" on each volume.

- **Bootable NVMe and VROC Consideration:** Although not directly relevant to SATA RAID, if the X9QRI-F+ is used with NVMe drives via PCIe adapters, Intel's RSTe 4.5 introduced a concept of NVMe RAID (requiring a feature called VROC on newer platforms, which X9 does not support since that's a feature of newer Xeon Skylake-SP platforms). So on X9, NVMe drives (if any) would just be standalone or use software RAID from the OS – they cannot be part of RSTe RAID. RSTe on X9 strictly covers the chipset's SATA ports.

- **UEFI BIOS Updates:** It's advisable to run a relatively recent BIOS on the X9QRI-F+ to ensure RAID stability. Supermicro's later BIOS releases for X9 boards often included fixes for device compatibility. For instance, if larger SATA drives (like 10+ TB) are used, updated firmware might handle their metadata better.

- **Interactions with IPMI/BMC:** The X9QRI-F+ has an IPMI management controller (the "-F" in the name). Some servers can propagate storage alerts to the BMC (e.g., through BIOS generating SEL events). It's not explicitly stated for X9, but keep the IPMI firmware updated as well – in case of drive failures, you would primarily rely on the OS/driver alert, but the BMC might also log an event if configured. There is also a header for an "alarm LED" or front panel in Supermicro chassis that might blink on RAID degradation if the board is wired for it via SGPIO – RSTe does support SGPIO enclosure management [28], which the backplane might use to drive LEDs. Just be aware that some trial-and-error might be needed to see if your chassis LEDs reflect the RAID status (often they do, for example, a failed drive might show a red LED if the backplane is compatible).

In short, the configuration of RSTe on the X9 platform is straightforward: enable RAID in BIOS, configure arrays, install drivers, and ensure your OS environment has the prerequisites (like .NET). Following these notes will help avoid common pitfalls (like missing drivers or non-booting volumes) and ensure a smooth integration of the Intel RSTe RAID into the system.

9. Documentation and Support Resources

For more information and official guidance, refer to the following resources:

- **Intel Download Page for RSTe 4.5.10.1021:** Intel's official download and description for this driver package is available on their website [60] [18]. It provides the release notes, supported hardware list, and the driver download itself (as *intel_rste_ahci_win_drivers_gui_cli_cim_4.5.10.1021.zip*). The included README (accessible via that page) details installation instructions and the list of fixes/updates in each version.

- **Intel RSTe User's Guide:** Although written for an earlier release, Intel has a **User's Guide for the C600 RSTe RAID** (July 2011, Order Number G40440-001). This manual covers the fundamentals of RSTe operation, RAID configuration, and features [61] [9]. Many concepts in that guide apply to 4.5 as well, including how to set up arrays in the option ROM and how to use the GUI/CLI. Supermicro's own user manual for the X9QRI-F+ briefly touches on enabling RAID in BIOS, but the Intel guide is more detailed on usage.

- **Release Notes / Readme:** The README text for version 4.5.10.1021 (named *rste_windows_drv_v4.5.10.1021_readme.txt*) contains the official list of fixes, known issues, and installation steps. Key points from it have been summarized in this report, but it can be referenced for any obscure issues. (For example, it enumerates the known restrictions we listed and credits some fixes to specific version updates.) Intel's site or mirror has this documentation [24] [40].

- **Security Advisory:** Intel's Security Center advisory INTEL-SA-00231 is the official statement on the ASM privilege escalation issue [43] [44]. It's a useful reference if one needs to understand the impact or decide on mitigation for that vulnerability in an environment using RSTe 4.5.

- **Supermicro Support:** Supermicro's support page for the X9QRI-F+ (archived, since the board is EOL) may offer driver downloads as well. Often, Supermicro reposts the Intel RSTe driver under their site for convenience. In case of any compatibility concerns or BIOS updates related to RAID, Supermicro's support resources or forums (like forums.supermicro.com) can be insightful.

- **Community Forums:** Enthusiast and admin communities such as the **Win-Raid Forum** and **ServeTheHome Forums** have threads discussing Intel RSTe behavior, performance tuning, and user experiences [62] [63]. These can be helpful to troubleshoot non-obvious issues (for instance, if trying to use consumer RST drivers on a server board, which is generally not recommended as noted by community experts).

By consulting the above documentation, hardware integrators and system administrators can get authoritative information straight from the source and find support for deploying the Intel RSTe RAID driver on the Supermicro X9QRI-F+ platform. The combination of Intel's official docs and Supermicro's platform knowledge will ensure a successful and stable integration of this RAID solution.

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